Hw 9

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4.1–1: Find constant c so f is a joint pmf

(a) $f(x,y) = c(x+2y),\ x = 1,2,\ y = 1,2,3.$

$$\sum_{x=1}^{2}\sum_{y=1}^{3}(x+2y) = \big(3+5+7\big) + \big(4+6+8\big) = 15+18 = 33.$$

Hence $c \cdot 33 = 1$, so

$$\boxed{c = \frac{1}{33}.}$$

(b) $f(x,y) = c(x+y),\ x = 1,2,3,\ y = 1,\ldots,x.$

Compute total sum:

$$x = 1:\ 2; \quad x = 2:\ 3+4 = 7; \quad x = 3:\ 4+5+6 = 15.$$

Total $= 2+7+15 = 24$, so

$$\boxed{c = \frac{1}{24}.}$$

(c) $f(x,y) = c$ for integer x, y with $6 \le x+y \le 8$ and $0 \le y \le 5$.

For each $y = 0,1,2,3,4,5$ the allowable x are $6-y, 7-y, 8-y$ (three values). So there are $6 \cdot 3 = 18$ points. Thus

$$\boxed{c = \frac{1}{18}.}$$

(d) $f(x,y) = c\left(\frac{1}{4}\right)^x \left(\frac{1}{3}\right)^y,\ x,y = 1,2,\ldots.$

The sum factorizes:

$$\sum_{x=1}^{\infty}\left(\tfrac{1}{4}\right)^x = \frac{\frac{1}{4}}{1-\frac{1}{4}} = \tfrac{1}{3}, \qquad \sum_{y=1}^{\infty}\left(\tfrac{1}{3}\right)^y = \frac{\frac{1}{3}}{1-\frac{1}{3}} = \tfrac{1}{2}.$$

Product $= \frac{1}{6}$, so $c \cdot \frac{1}{6} = 1$ and

$$\boxed{c = 6.}$$

4.1–3: $f(x,y) = \dfrac{x+y}{32},\ x = 1,2,\ y = 1,2,3,4$

(a) Marginal pmf of X:

$$f_X(x) = \sum_{y=1}^{4} \frac{x+y}{32} = \frac{4x + \sum_{y=1}^{4} y}{32} = \frac{4x+10}{32} = \frac{2x+5}{16}.$$

So

$$\boxed{f_X(1) = \tfrac{7}{16}, \qquad f_X(2) = \tfrac{9}{16}.}$$

(b) Marginal pmf of Y:

$$f_Y(y) = \sum_{x=1}^{2} \frac{x+y}{32} = \frac{3+2y}{32}.$$

So

$$\boxed{f_Y(1) = \tfrac{5}{32},\ f_Y(2) = \tfrac{7}{32},\ f_Y(3) = \tfrac{9}{32},\ f_Y(4) = \tfrac{11}{32}.}$$

(c) $P(X > Y)$. The only pair with $X > Y$ is $(2,1)$, so

$$P(X > Y) = f(2,1) = \frac{2+1}{32} = \boxed{\tfrac{3}{32}.}$$

(d) $P(Y = 2X)$. Pairs: $(1,2), (2,4)$. So

$$P = \frac{1+2}{32} + \frac{2+4}{32} = \frac{3}{32} + \frac{6}{32} = \boxed{\tfrac{9}{32}.}$$

(e) $P(X + Y = 3)$. Pairs: $(1,2), (2,1)$.

$$P = \frac{3}{32} + \frac{3}{32} = \boxed{\tfrac{3}{16}.}$$

(f) $P(X \le 3 - Y)$ i.e. $X + Y \le 3$. Valid pairs: $(1,1), (1,2), (2,1)$.

$$P = \frac{2}{32} + \frac{3}{32} + \frac{3}{32} = \frac{8}{32} = \boxed{\tfrac{1}{4}.}$$

(g) Independence? No. Example:

$$f(1,1) = \frac{1+1}{32} = \frac{2}{32} \neq f_X(1) f_Y(1) = \frac{7}{16} \cdot \frac{5}{32} = \frac{35}{512}.$$

So X and Y are dependent.

(h) Means and variances.

Compute

$$E[X] = 1 \cdot \frac{7}{16} + 2 \cdot \frac{9}{16} = \frac{25}{16}.$$

$$E[X^2] = 1^2 \cdot \frac{7}{16} + 4 \cdot \frac{9}{16} = \frac{43}{16}, \qquad \mathrm{Var}(X) = \frac{43}{16} - \left(\frac{25}{16}\right)^2 = \frac{63}{256}.$$

$$E[Y] = \sum_{y=1}^{4} y \cdot \frac{3+2y}{32} = \frac{90}{32} = \frac{45}{16}.$$

$$E[Y^2] = \frac{290}{32} = \frac{145}{16}, \qquad \mathrm{Var}(Y) = \frac{145}{16} - \left(\frac{45}{16}\right)^2 = \frac{295}{256}.$$

Thus

$$\boxed{E[X] = \tfrac{25}{16},\ \mathrm{Var}(X) = \tfrac{63}{256}; \qquad E[Y] = \tfrac{45}{16},\ \mathrm{Var}(Y) = \tfrac{295}{256}.}$$

4.1–10: Smoking survey (multinomial, $n = 7$)

Preferences: $p_1 = 0.78$ (nonsmokers), $p_2 = 0.01$ (smokers), $p_3 = 0.21$ (don't care). Let $X = \#$ preferring nonsmokers, $Y = \#$ preferring smokers.

(a) Joint pmf (trinomial/ multinomial):

$$\boxed{P(X = i, Y = j) = \frac{7!}{i!\, j!\, (7-i-j)!}\, 0.78^{\,i}\, 0.01^{\,j}\, 0.21^{\,7-i-j},}$$

for integers $i \geq 0,\ j \geq 0,\ i + j \leq 7$ (support: $i + j \leq 7$).

(b) Marginal pmf of X:

$$X \sim \mathrm{Binomial}(7, 0.78), \qquad \boxed{P(X = i) = \binom{7}{i} 0.78^{\,i} (0.22)^{\,7-i},\ i = 0, \ldots, 7.}$$

(Here $0.22 = 1 - 0.78$ collects the other two categories.)

4.2–1: (Same joint as 4.1–3) compute means, variances, covariance, correlation

From previous results:

$$E[X] = \frac{25}{16}, \qquad E[Y] = \frac{45}{16},$$
$$\mathrm{Var}(X) = \frac{63}{256}, \qquad \mathrm{Var}(Y) = \frac{295}{256}.$$

Compute

$$E[XY] = \sum_{x=1}^{2} \sum_{y=1}^{4} xy \frac{x+y}{32} = \frac{140}{32} = \frac{35}{8}.$$

Hence

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y] = \frac{35}{8} - \frac{25}{16} \cdot \frac{45}{16} = -\frac{5}{256}.$$

Correlation:

$$\rho = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}} = \frac{-5/256}{\sqrt{(63/256)(295/256)}} = -\frac{5}{\sqrt{63 \cdot 295}} \approx -0.0367.$$

Independence? No (joint $\neq$ product), covariance small but nonzero.

4.2–4: Trinomial with $n = 3$, $p_X = \frac{1}{6}$, $p_Y = \frac{1}{2}$

Use standard multinomial marginal formulas:

$$E(X) = np_X = 3 \cdot \frac{1}{6} = \boxed{\tfrac{1}{2}}, \qquad E(Y) = np_Y = 3 \cdot \frac{1}{2} = \boxed{1.5}.$$

$$\mathrm{Var}(X) = np_X(1 - p_X) = 3 \cdot \frac{1}{6} \cdot \frac{5}{6} = \boxed{\tfrac{5}{12}},$$

$$\mathrm{Var}(Y) = np_Y(1 - p_Y) = 3 \cdot \frac{1}{2} \cdot \frac{1}{2} = \boxed{\tfrac{3}{4}}.$$

$$\mathrm{Cov}(X, Y) = -np_Xp_Y = -3 \cdot \frac{1}{6} \cdot \frac{1}{2} = \boxed{-\tfrac{1}{4}}.$$

$$\rho = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}} = -\frac{1/4}{\sqrt{\frac{5}{12} \cdot \frac{3}{4}}} = -\frac{1}{\sqrt{5}} \approx -0.4472.$$

4.2–9: $f(x, y) = \frac{1}{4}$ on $S = \{(0, 0), (1, 1), (1, -1), (2, 0)\}$

(a) Independence? No. Example:

$$f(1, 1) = \tfrac{1}{4} \neq f_X(1)f_Y(1) = \tfrac{1}{2} \cdot \tfrac{1}{4} = \tfrac{1}{8}.$$

(b) Compute moments:

$$E[X] = 0 \cdot \tfrac{1}{4} + 1 \cdot \tfrac{1}{2} + 2 \cdot \tfrac{1}{4} = 1, \qquad E[Y] = 0 \cdot \tfrac{1}{2} + 1 \cdot \tfrac{1}{4} + (-1) \cdot \tfrac{1}{4} = 0.$$

$$E[XY] = 0 + \tfrac{1}{4} - \tfrac{1}{4} + 0 = 0.$$

So

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0 - 1 \cdot 0 = 0.$$

Variances:

$$E[X^2] = 0 \cdot \tfrac{1}{4} + 1 \cdot \tfrac{1}{2} + 4 \cdot \tfrac{1}{4} = \tfrac{3}{2} \quad \Rightarrow \quad \mathrm{Var}(X) = \tfrac{3}{2} - 1^2 = \tfrac{1}{2}.$$

$$E[Y^2] = 0 \cdot \tfrac{1}{2} + 1 \cdot \tfrac{1}{4} + 1 \cdot \tfrac{1}{4} = \tfrac{1}{2} \quad \Rightarrow \quad \mathrm{Var}(Y) = \tfrac{1}{2} - 0^2 = \tfrac{1}{2}.$$

Thus correlation

$$\boxed{\mathrm{Cov}(X, Y) = 0, \qquad \rho = 0.}$$

(Dependent variables can have covariance/correlation zero.)